# ANNEXURE - II

## SKILL ENHANCEMENT COURSES

## SEMESTERS I, II and III

## COMMON FOR ALL B.VOC PROGRAMMES

# SEMESTER I

| S. No. | Course Code | Course Title |
|---|---|---|
| 1. | B.Voc. SEC 101 | Basics of Fashion Designing |
| 2. | B.Voc. SEC 102 | Applied Hindi in Translation and Journalism |
| 3. | B.Voc. SEC 103 | Music in Daily Life |
| 4. | B.Voc. SEC 104 | Business and Office Communication |
| 5. | B.Voc. SEC 105 | Photo Editing with Photoshop/PhotoPEA |
| 6. | B.Voc. SEC 106 | Basics of Excel |
| 7. | B.Voc. SEC 107 | Digital Art Using Photoshop |

**SKILL ENHANCEMENT COURSES ADOPTED FROM OTHER FACULTIES**

| S. No. | Course Title |
|---|---|
| 1. | Data Analytics* |
| 2. | Data Collection and Handling Techniques** |
| 3. | Biosafety*** |
| 4. | Digital Marketing† |
| 5. | Security Analysis†† |
| 6. | Creative Writing††† |

**Note:** *Adopted from B.COM/BBA [BCOM/BBA (SEC – 101A)]
**Adopted from BA(Economics) [ECO (SEC – 1)]
***Adopted from Biotechnology [BIOT (SEC – 1T)]
†Adopted from B.COM/BBA [BCOM/BBA (SEC – 101)]
††Adopted from B.COM/BBA [BCOM/BBA (SEC) – 101B)]
†††Adopted from Journalism & Mass Communication SEC Semester-II [SCS (SEC) – 103]

| | COURSE TITLE | BASICS OF FASHION DESIGNING |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 101 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 90 hours of practical (6 hours/week of practical)<br>(1 credit = 30 hours of practical based on 6 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits |
| 5 | **Max. Marks and Time** | 75 Marks (60 PR+ 15 IA)<br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. |
| 6 | **Course Objectives** | i. To familiarize the students with the terminologies related to fashion and construction.<br>ii. To train the students in applying the art and aesthetic in designing and construction of clothing and accessories.<br>iii. To create awareness about latest trends in the fashion industry.<br>iv. To impart practical knowledge about skill of designing and construction of dresses.<br>v. To develop the skills of selecting effective dress and accessories as per<br>vi. different figure types, occasions, age groups and seasons.<br>vii. To provide insight about the Indian Heritage- traditional embroideries and printing techniques.<br>viii. To enable the students to engage in creative skills of sketching, tie and dye and computer aided designing. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br>CO1. Join fashion industry as a designer<br>CO2. Establish manufacturing and readymade garments units<br>CO3. Own business/boutique<br>CO4. Join Fashion accessories house<br>CO5. Join/run a fashion house<br>CO6. Be a fashion merchandiser<br>CO7. Be a accessories designer<br>CO8. Be a fashion stylist<br>CO9. Be a textile designer<br>CO10. Be a teacher/instructor |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 10 | **Instructions for paper setters and candidates** | **The syllabus has been divided into TWO units**:<br>For practical exam, there should be at least two questions for practical performance of 50 marks and one written part of maximum 10 marks. |

11 **Course Content**

**UNIT-I**

Introduction to sewing equipment, Sewing machine and its parts.
Maintenance and care of sewing machine.
Drafting of Child's Bodice Block and Adult Bodice Block including Basic sleeve.
A project on market survey to understand the latest trends in Fabric Trimming
Visual identification of various fibres, burning test of fibres

**UNIT-II**

A study of different textures of fabric by collecting the samples.
Creation of various textures using different mediums.
Seam and seam finishes- French, run and fell, pinking, Mantua, Counter hem.
Basting and hemming
Facing and binding
Plackets and fasteners
Fullness treatment – gathers, pleats and shirring
Sketching of basic fashion figure
Sketching of nose, ears, lips, face and hairstyle.

12 **Suggested Resources for Reading**


1. Dorothy Wood, The Practical Encyclopedia of Sewing, 1999, Lorenz Books, Anness Publishing Ltd., New York.
2. Complete Book of Sewing- A Practical step-by-step Guide to Sewing Techniques, 1996, Dorling Kindersley Publishing Inc., London.
3. Sewing for the Apparel Industry- Claire Shaeffer Prentice Hall, (2011).
4. Helen Joseph Armstrong, Pattern Making for Fashion Design, 2000, Dorling Kindersley (India) Pvt. Ltd. India
5. Paine Sherla, "Embroidered Textiles", Thames and Hudson Ltd, 1990
6. Chattopadhyay K, "Indian Embroidery", Wiley Eastern Ltd., New Delhi, 1977


**Note**: Latest edition of text books may be used.

<table>
<tr><th></th><th>COURSE TITLE</th><th>APPLIED HINDI IN TRANSLATION AND JOURNALISM</th></tr>
<tr><td>1</td><td>Course Code</td><td>B.Voc. SEC 102</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Skill Enhancement Courses (SEC)</td></tr>
<tr><td>3</td><td>Course</td><td>30 hours of lectures for Theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>3 credits</td></tr>
<tr><td>5</td><td>Max. Marks and Time</td><td>75 Marks
<table>
<tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr>
<tr><td>40</td><td>10</td><td>20</td><td>5</td></tr>
</table>
i. The exam of this course will be of 3hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of thedevelopments in the subject up to 6 months before the examination.</td></tr>
<tr><td>6</td><td>Course Objectives</td><td>The objective of this course is to introduce the learners to skillful applied Hindi in accordance with the National Education Policy- 2020. The main objective is to make the student proficient in all the four skills of the language so that he can get employment in the future by choosing possible opportunities for him or herself in the field of language.</td></tr>
<tr><td>7</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to:<br>CO1. The students will be able to become aware of masscommunication, journalism and media.<br>CO2. The program emphasizes skill development in the field oftranslation.<br>CO3. This program will develop creative writing skills in the students.<br>CO4. The students will develop practical knowledge related to news writing, feature writing, report writing.<br>CO5. By completing this program, the learners will get various new employment opportunities as per the need of today along with the traditional employment opportunities.</td></tr>
<tr><td>8</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Field experience, Interactive sessions, PPT, Practical and Group Discussions, hands on training etc.</td></tr>
<tr><td>9</td><td>Instructions for paper setters and candidates</td><td>The syllabus has been divided into TWO units:<br>Note: The question paper of each subject covering the entire course shall be divided into three sections<br>Section A (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks.<br>Section B (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.<br>Section C (15 marks) This section will consist of essay type/ numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.</td></tr>
</table>

10 **Course Content**

UNIT-I

**अनुवाद :** अर्थ, प्रकार, सामान्य सिद्धांत, आधुनिक समय में अनुवाद में इंटरनेट की भूमिका।

**सृजनात्मक लेखन :** कहानी का सामान्य परिचय व लेखन कला, नाटक का सामान्य परिचय व लेखन कला।

**हिंदी ब्लॉग लेखन व ऑनलाइन प्रकाशन :** कंप्यूटर पर हिंदी लेखन और प्रकाशन, हिंदी ब्लॉग लेखन कला, हिंदी ई- पत्र पत्रिकाओं का सामान्य परिचय।

UNIT-II

**मीडिया लेखन :** मीडिया में हिंदी, रिपोर्ट का सामान्य परिचय व लेखन कला, फीचर का सामान्य परिचय व लेखन कला, समाचार का सामान्य परिचय व लेखन कला।

**जनसंचार और मीडिया :** जनसंचार और मीडिया, जनसंचार के माध्यम, मीडिया और हिंदी भाषा।

**रेडियो :** परिचय, विविध कार्यक्रम, रेडियो समाचार, वाचन की सावधानियाँ, आधुनिक समय में रेडियो से पॉडकास्ट की यात्रा।

**दूरदर्शन :** परिचय, प्रस्तुतिकरण के क्षेत्र, विविध कार्यक्रम, दूरदर्शन के विज्ञापनों की भाषा, दूरदर्शन पटकथा लेखन।

11 **Suggested Resources for Reading**


1. अनुवाद प्रक्रिया और स्वरुप : कैलाशचंद्र भाटिया
2. बदलता समाज मनोविज्ञान और हिंदी: पूरनचंदन टंडन
3. मीडिया कालीन हिंदी : स्वरूप एंव सम्भावनाएँ : डा अर्जुन चव्हाण
4. नए जन संचार माध्यम और हिंदी - डॉ. सुधीश पचौरी
5. समाचार एवं फीचर लेखन - प्रो. डॉ. संजीव भानावत
6. डिजिटल ब्राडकास्टिंग जर्नलिज्म : जितेन्द्र शर्मा
7. कम्प्यूटर और हिंदी : हरि मोहन


**Note: Latest edition of text books may be used.**

| | COURSE TITLE | MUSIC IN DAILY LIFE |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 103 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 15 hours of lectures for Theory + 60 hours of practicals<br>(1 credit = 15 hours of theory learning based on 1 hrs./week)<br>(1 credit = 30 hours of practical based on 4 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits [1+2] |
| 5 | **Max. Marks and Time** | 75 Marks<br><table><tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr><tr><td>20</td><td>5</td><td>40</td><td>10</td></tr></table><br>i. The exam of this course will be of 3hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. |
| 6 | **Course Objectives** | To inculcate in learner's various aspects of music and make it an essential part of their day-to-day life for mental, physical, emotional fitness and enable them to use nuances of music in their profession, especially media and entertainment. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br>CO1. Understand basic music terminologies, instruments, tempo, rhythm and apply them in their routine life.<br>CO2. Imbibe positive values such as confidence, independence and self-perseverance.<br>CO3. Use music as a healing therapy to reduce stress in today's fast paced life.<br>CO4. Choose from various dimensions of music as a career (singer, music director etc). |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into TWO units**.<br>**Note**: The question paper of each subject covering the entire course shall be divided into three sections<br>**Section A** (4 marks) This section will have 4 short answer questions from the entire syllabus. Students are required to attempt 2 questions from this section. Each question will carry 2 marks; the total weightage being 4 marks.<br>**Section B** (8 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks.<br>**Section C** (8 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks. |

10 **Course Content**

**UNIT-I**

Basic Knowledge of Indian Music

Elementary knowledge of Swar, Alankar, Laya and basic taalas in Music.
Indian Music and different forms.
Music and Health: Stress buster, relief from physical & Emotional fatigue.

**UNIT-II**

Practical knowledge and Training

Basic training of Tabla and Harmonium/keyboard.
Basic training of Karaoke using different Music Apps.
Practical training of the following: Geet, Gazal, Sufi, Traditional and Folk Music.
Income generating through Music: Music Tuitions, Teaching, Production house/studio and Musician etc.

11 **Suggested Resources for Reading**


1. Bhartiya Sangeet aur Manovigyan, Dr Vasudha Kulkarni
2. Bhartiya Sangeet Vaadya, Dr. Lal Mani Mishra
3. Nibandh Sangeet, Dr. Lakshmi Narain Garg
4. Sangeet Evam Manovigyan, Dr. Kiran Tiwari
5. Sangeet Kala Ka Itihas, Dr. Panna Lal Madan
6. Sangeet Shastra Vigyan, Dr. Panna Lal Madan
7. Sangeet SAR Part I, II & III, Mrs. Veena Mankaran


**Note**: Latest edition of text books may be used.

<table>
<tr><th></th><th>COURSE TITLE</th><th>BUSINESS AND OFFICE COMMUNICATION</th></tr>
<tr><td>1</td><td>Course Code</td><td>B.Voc. SEC 104</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Skill Enhancement Courses (SEC)</td></tr>
<tr><td>3</td><td>Course Duration/ LP (Lectures, Practical)</td><td>30 hours of lectures for Theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>3 credits</td></tr>
<tr><td>5</td><td>Max. Marks and Time</td><td>75 Marks<br><table><tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr><tr><td>40</td><td>10</td><td>20</td><td>5</td></tr></table><br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.</td></tr>
<tr><td>6</td><td>Course Objectives</td><td>i. To acquire core competency of Fundamentals of Communication & Correspondence in Office & Business houses<br>ii. To use electronic media for Office and Business Communication</td></tr>
<tr><td>7</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to:<br>CO1: Understand fundamentals of communication and correspondence being followed in a Modern Office<br>CO2: Understand the use of electronic media for office and business communication</td></tr>
<tr><td>8</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>9</td><td>Instructions for paper setters and candidates</td><td>The syllabus has been divided into TWO units.<br>Note: The question paper of each subject covering the entire course shall be divided into three sections<br>Section A (8 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2 marks; the total weightage being 8 marks.<br>Section B (16 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks.<br>Section C (16 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks. Will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks.</td></tr>
</table>

10 **Course Content**

**UNIT-I**

Office Communication: Meaning, Importance and Process of Communication in Office; Objectives of Effective Communication in Office; Forms of Communication-Oral, Written, Visual and Audio-Visual; Types of Communication (Personal, Business and Official); Barriers to Communication, Overcoming the Barriers
Office Correspondence: Types of Official Letters: Official Letter, Demi-Official Letter, Office Order, Memorandum/Memo Letters, Notification, Endorsement, Inter-departmental Communication; Report Writing: Types, Characteristics, Importance, Elements, Structure, Process of Writing, Order of Writing, The Final Draft; Presentation: Importance, Characteristics, Presentation Plan, Power Point Presentation, Visual Aids

**UNIT-II**

Business Correspondence: Electronic Tele-Conferencing, Video-Conferencing, Computer, Printer, Scanner, Internet, E-mail, Social Media, Telephone, Fax; Written Letter of Enquiry and Reply, Inviting Quotations, Sending Quotations, Placing Order, Confirmation Letter, Modification and Non-Acceptance of Orders, Letters of Complaint and Adjustments, Circular Letters
Office Meetings: Notices, Agenda, Resolution, Minutes and Confirmation of the Minutes

11 **Suggested Resources for Reading**


1. Lesikar, R.V. & Flatley, M.E.; Basic Business Communication Skills for Empowering the Internet Generation, Tata McGraw Hill Publishing Company Ltd. New Delhi.
2. Bovee, and Thill, Business Communication Today, Pearson Education
3. Shirley Taylor, Communication for Business, Pearson Education
4. Locker and Kaczmarek, Business Communication: Building Critical Skills, TMH


**Note**: Latest edition of text books may be used.

| | COURSE TITLE | PHOTO EDITING WITH PHOTOSHOP/PHOTOPEA |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 105 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 30 hours of lectures for Theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits |
| 5 | **Max. Marks and Time** | 75 Marks<br><table><tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr><tr><td>40</td><td>10</td><td>20</td><td>5</td></tr></table><br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. |
| 6 | **Course Objectives** | The objective of this course is to familiarize students with the fundamentals of Photoshop to design, edit and manipulate images for logos, brochures and web development. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br>CO1. Editing Images.<br>CO2. Enhance and manipulate captured images.<br>CO3. Create logos, brochures etc.<br>CO4. Earn skills to work as a freelance image editor. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into TWO units**:<br>**Note**: The question paper of each subject covering the entire course shall be divided into three sections.<br>**Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks.<br>**Section B** (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.<br>**Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. |

10 **Course Content**

**UNIT-I**

Photoshop- Feature and advantages, User Interface, file opening and saving, file formats, Palettes, Image size, resolution setting, understanding color modes and models, color model conversions, color correction, foreground & background colors, color picker, color swatches, eyedropper

Selection Tools- Marquee, Lasso, Magic Wand and selecting by color range, Modifying selections, Selection Tool Palette and Interface.

Painting tools- Brush, pencil, eraser, magnetic eraser, Background eraser, art history tool, History Brush, Brush Tool Palette and Interface, Rotating, Scaling, and Distorting with Transformations

**UNIT-II**

Layers - Using Multiple Layer - Composition effects - Re-arranging and stacking order - Layer effects. Layers with Masks and Channels - Using Quick Mask - Saving Selections - Understand Color Channel vs. Channel Mask - Layer Mask - Clipping Mask. Healing tool, patch tool & clone stamp. Text tool.

Image slicing, Image cropping, Retouching Images – blur, sharpen, smudge, burn, sponge& dodge. Image filters. Integrating Vector Graphics - Shape Tools - Pen Tools & Selection Tools - Paths and Clipping Paths – Image adjustments

11 **Suggested Resources for Reading**


1. eBook - Adobe Photoshop Help https://helpx.adobe.com/pdf/photoshop_reference.pdf
2. Adobe Photoshop Classroom in a Book 2024 Release: The Official Training Workbook from Adobe, by Conrad Chavez
3. Adobe Photoshop CC Classroom in a Book, Pearson Edition (2018).
4. Introduction to Photoshop: An Essential Guide for Absolute Beginners Paperback – Import, 2022 by Kevin Wilson, APress
5. The Photoshop Toolbox: Essential Techniques for Mastering Layer Masks, Brushes, and Blend Modes Paperback – 31 October 2018, by Glyn Dewis, Rockynook
6. The Photoshop Layers and Selections Workshop Paperback – Import, 29 June 2021, by Glyn Dewis, Rockynook
7. https://www.photopea.com/
8. https://www.photopea.com/tuts/


**Note**: Latest edition of text books may be used.

<table>
<tr><th></th><th>COURSE TITLE</th><th>BASICS OF EXCEL</th></tr>
<tr><td>1</td><td>Course Code</td><td>B.Voc. SEC 106</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Skill Enhancement Courses (SEC)</td></tr>
<tr><td>3</td><td>Course Duration/ LP (Lectures, Practical)</td><td>15 hours of lectures for Theory + 60 hours of practical<br>(1 credit = 15 hours of theory learning based on 1 hrs./week)<br>(1 credit = 30 hours of practical based on 4 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>3 credits (1 Theory + 2 Practical)</td></tr>
<tr><td>5</td><td>Max. Marks and Time</td><td>75 Marks<br><table><tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr><tr><td>20</td><td>5</td><td>40</td><td>10</td></tr></table><br>i. The exam of this course will be of 3hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.<br>iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>6</td><td>Course Objectives</td><td>The main objective of the course is to make students learn about managing data using spreadsheets in EXCEL.</td></tr>
<tr><td>7</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to:<br><br>CO1. Create, store and print data.<br>CO2. Sort and filter data.<br>CO3. Manage data in different formats.<br>CO4. Apply formulae and functions on data.<br>CO5. Visualize data using charts.</td></tr>
<tr><td>8</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>9</td><td>Instructions for paper setters and candidates</td><td>The syllabus has been divided into TWO units:<br><br>Note: The question paper of each subject covering the entire course shall be divided into three sections.<br>Section A (4 marks) This section will have 4 short answer questions from the entire syllabus. Students are required to attempt 2 questions from this section. Each question will carry 2 marks; the total weightage being 4 marks.<br>Section B (8 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks.<br>Section C (8 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks.</td></tr>
</table>

10 **Course Content**

**UNIT-I**

Learning Anatomy of Excel: Understanding Workbook, Worksheet, Title bar, Ribbon, Tabs, Groups, Row, Column, Cells, Name Box and Formula Bar; Quick access Tool bar, Status bar, File Tab and Back stage view, scroll bars, Zoom control; Creating, opening, saving and closing workbooks

Adding and Editing Data: Textual, numeric, currency and date data; Using cut, copy, paste and paste special operations, Using find and replace feature, using formulae, Relative, absolute, and mixed referencing; Deactivating automatic changes

Formatting Cells: Using Format Painter; Understanding and using Number formats, Currency Formats, Date and Time formats, Font formats, Alignment, Borders; Using conditional formatting; Autofitting Columns & Rows, Hiding/Unhiding Columns & Rows, Inserting/Deleting Cells, Columns & Rows; freezing rows and columns, splitting screen, Wrapping Text and merging cells; Understanding and using Range, Naming Range

**UNIT-II**

Working with multiple Worksheets: Inserting new worksheet, copying, moving, deleting, and renaming worksheets; understanding worksheet view options

Printing Worksheet: Inserting page breaks, setting Up Print Area, customizing Headers & Footers, adding page numbers, using Page setup options, Print Titles –Repeat Rows / Columns

Managing data: Creating, sorting and filtering a database, Using custom sorting

Managing data using Functions: Math, statistical, String/Text, Date and Time functions

Understanding Charts: Chart types, Creating Embedded Charts, formatting Charts: Changing Chart types, adding Titles, Legends and Gridlines, Printing Charts

11 **Suggested Resources for Reading**


1. Kabir Das, Microsoft Excel: Shortcut keys and Formulas, Paperback
2. Manish Nigam, Data Analysis with Excel, Paperback, BPB Publications
3. L. Winston Wayne, Microsoft Excel 2019: Data Analysis & Business Model Paperback


**Note**: Latest edition of text books may be used.

<table>
<tr><th></th><th>COURSE TITLE</th><th>DIGITAL ART USING PHOTOSHOP</th></tr>
<tr><td>1</td><td>Course Code</td><td>B.Voc. SEC 107</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Skill Enhancement Course (SEC)</td></tr>
<tr><td>3</td><td>Course Duration/ LP (Lectures, Practical)</td><td>15 hours of lectures for Theory + 60 hours of practicals<br>(1 credit = 15 hours of theory learning based on 1 hrs./week)<br>(1 credit = 30 hours of practical based on 4 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>3 credits [1+2]</td></tr>
<tr><td>5</td><td>Max. Marks and Time</td><td>75 Marks<br><table><tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr><tr><td>20</td><td>5</td><td>40</td><td>10</td></tr></table><br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.</td></tr>
<tr><td>6</td><td>Course Objectives</td><td>The objective of this course is to familiarize students with the fundamentals of Photoshop to design, edit and manipulate images for logos, brochures and web development.</td></tr>
<tr><td>7</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to:<br>CO1. Editing, enhancing and manipulating captured images.<br>CO2. Create logos, Cards, Brochures.<br>CO3. Work as a freelance Digital Artist.</td></tr>
<tr><td>8</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>9</td><td>Instructions for paper setters and candidates</td><td>The syllabus has been divided into TWO units:<br>Note: The question paper of each subject covering the entire course shall be divided into three sections<br>Section A (4 marks) This section will have 4 short answer questions from the entire syllabus. Students are required to attempt 2 questions from this section. Each question will carry 2 marks; the total weightage being 4 marks.<br>Section B (8 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks.<br>Section C (8 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks.</td></tr>
</table>

10 **Course Content**

**UNIT-I**

Photoshop- Feature and advantages, User Interface, file opening and saving, file formats, Pallets, Image size, resolution setting, understanding color modes and models, color model conversions, color correction, foreground & background colors, color picker, color swatches, eyedropper.

Selection Tools- Marquee, Lasso, Magic Wand and selecting by color range, Modifying selections, selection Tool Palette and Interface. Painting tools- Brush, pencil, eraser, magnetic eraser, Background eraser, art history tool, History Brush, Brush Tool Palette and Interface, Rotating, Scaling, and Distorting with Transformations

**UNIT-II**

Layers - Using Multiple Layer, Composition effects, Re-arranging and stacking order, Layer effects. Layers with Masks and Channels - Using Quick Mask, Saving Selections, Understand Color Channel vs. Channel Mask - Layer Mask, Clipping Mask. Healing tool, patch tool & clone stamp, Text tool.

Image slicing, Image cropping, Retouching Images – blur, sharpen, smudge, burn, sponge& dodge. Image filters. Integrating Vector Graphics -Shape Tools - Pen Tools & Selection Tools - Paths and Clipping Paths – Image adjustments

11 **Suggested Resources for Reading**


1. eBook - Adobe Photoshop Help https://helpx.adobe.com/pdf/photoshop_reference.pdf
2. Adobe Photoshop Classroom in a Book 2024 Release: The Official Training Workbook from Adobe, by Conrad Chavez
3. Adobe Photoshop CC Classroom in a Book, Pearson Edition (2018).
4. Introduction to Photoshop: An Essential Guide for Absolute Beginners Paperback – Import, 2022 by Kevin Wilson, APress
5. The Photoshop Toolbox: Essential Techniques for Mastering Layer Masks, Brushes, and Blend Modes Paperback – 31 October 2018, by Glyn Dewis, Rockynook
6. The Photoshop Layers and Selections Workshop Paperback – Import, 29 June 2021, by Glyn Dewis, Rockynook


**Note**: Latest edition of text books may be used.

# SEMESTER II

| S. No. | Course Code | Course Title |
|---|---|---|
| 1. | B.Voc. SEC 201 | Python Programming |
| 2. | B.Voc. SEC 202 | Entrepreneurial Skills in Microbiology |
| 3. | B.Voc. SEC 203 | Basics of Fashion Ornamentation |
| 4. | B.Voc. SEC 204 | Front Office Management |
| 5. | B.Voc. SEC 205 | Introduction to Econometrics |
| 6. | B.Voc. SEC 206 | GST Return Filing |

**SKILL ENHANCEMENT COURSES ADOPTED FROM OTHER FACULTIES**

| S. No. | Course Title |
|---|---|
| 1. | Data Analysis for Beginners* |
| 2. | Desktop Publishing using GIMP** |
| 3. | Biostatistics*** |
| 4. | E-Commerce† |
| 5. | Event Management†† |

**Note:** *Adopted from the Department of Economics [ECO (SEC – 1)]
** Adopted from BCA [BCA (SEC – 206)]
*** Adopted from Biotechnology [BIOT (SEC – 2T)]
†Adopted from B.COM/BBA [BCOM/BBA (SEC – 201)]
††Adopted from Journalism & Mass Communication SEC Semester-II [SCS (SEC – 153)]

<table>
<tr><th></th><th>COURSE TITLE</th><th>PYTHON PROGRAMMING</th></tr>
<tr><td>1</td><td>Course Code</td><td>B.Voc. SEC 201</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Skill Enhancement Courses (SEC)</td></tr>
<tr><td>3</td><td>Course Duration/ LP (Lectures, Practical)</td><td>30 hours of lectures for Theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>3 credits [2+1]</td></tr>
<tr><td>5</td><td>Max. Marks and Time</td><td>75 Marks<br><table><tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr><tr><td>40</td><td>10</td><td>20</td><td>5</td></tr></table><br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.<br>iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>6</td><td>Course Objectives</td><td>The objective of this course is to familiarize students with the fundamentals of Python programming. By the end of this course, students should be able to understand the basic syntax of Python, apply control flow structures for decision-making and looping, define and utilize functions effectively, and work with essential data structures such as lists, tuples, dictionaries, and sets. Additionally, students will gain proficiency in file handling.</td></tr>
<tr><td>7</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to:<br>CO1. Understand the core syntax and semantics of Python programming language.<br>CO2. develop Python programs with conditionals and loops.<br>CO3 understand the process of structuring the data using lists, dictionaries, tuples and sets.<br>CO4. work with the arrays, strings and functions<br>CO5. Understand file system and analyse the applications of Python in real life</td></tr>
<tr><td>8</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>9</td><td>Instructions for paper setters and candidates</td><td>The syllabus has been divided into TWO units:<br>Note: The question paper of each subject covering the entire course shall be divided into three sections.<br>Section A (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks.<br>Section B (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.</td></tr>
</table>

**Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks**.**

10 **Course Content**

**UNIT-I**

Introduction: Python Programming Language, History, features, Python Shell, Python Interpreter, Types of IDE

Data Types, Variables, operators and Other Basic Elements: variables, keywords, data types, Type conversion, operators and operands, Expressions, Precedence and Associativity, Indentation, comments, Print statement and basic input/output

Conditional Statements: if statement, if-else statement and elif statement, Looping: while loop, for loop, nested loops, Control statements: pass, continue, break.

Python Data structure- List- Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, Dictionary - Creating Dictionary, Accessing and Modifying key:value Pairs in Dictionaries, Built In Functions Used on Dictionaries, Dictionary Methods, Tuple and Set Creating Tuples, Basic Tuple Operations, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Tuple Methods, Sets, Set Methods, Traversing of Sets

**UNIT-II**

Array, String - String Basics: Creating and Storing, indexing, Slicing and Joining Strings, Basic String Operations, String Method.

Functions: Introduction to functions, Function definition and calling, Function parameters, Default argument function, Variable argument function, Return Values and return Statements, Recursive functions, built-in functions, scope of variables, Lambda function in python

Modules: Introduction, Importing modules, Working with standard library modules-Math Module

File handling: Concept of Files, File opening in various modes and closing of a file, Reading and writing functions, manipulating file pointer using seek, Creating and Reading Text Data, appending data to a file, closing files, Reading and Writing Binary Files, The Pickle Module, Reading and Writing CSV Files.

11 **Suggested Resources for Reading**


1. Gowrishankar S, Veena A, Introduction to Python Programming, CRC Press/Taylor & Francis
2. Kamthane, A. N., &Kamthane, A.A., Programming and Problem Solving with Python, McGraw Hill Education
3. Wesley J Chun, Core Python Applications Programming, Third Edition. Pearson Publication
4. Balaguruswamy E. Introduction to Computing and Problem-Solving using Python, McGraw Hill Education
5. Reema Thareja, Python Programming using problem solving approach, Oxford University Press.
6. Programming Python by Mark Lutz


**Note**: Latest edition of text books may be used.

| | COURSE TITLE | ENTREPRENEURIAL SKILLS IN MICROBIOLOGY |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 202 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 90 hours of practicals<br>(1 credit = 15 hours of theory learning based on 3 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits |
| 5 | **Max. Marks and Time** | 75 Marks (60 PR+ 15 IA)<br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. |
| 6 | **Course Objectives** | The course is designed to inculcate and strengthen the entrepreneurial skills of students in the field of microbiology. It also aims to apprise the students of the various bio-entrepreneurship opportunities in the field of microbiology and the step wise approach to initiate their own start-up. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br>CO1.Learn about various bio-entrepreneurship opportunities in the field of microbiology.<br>CO2. Strengthen the various microbiology-based skills that can be taken up as future entrepreneurial career options.<br>CO3.Learn the commercial aspects, requirements and approach required to start a start-up. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into two units:**<br>For practical exam, there should be at least two questions for practical performance of 50 marks and one written part of maximum 10 marks. |
| 10 | **Course Content** | **UNIT-I**<br>Sustainable Waste Management and Renewable Bioenergy Production<br>Process of spawn production and mushroom cultivation<br>Process of bioethanol production from waste substrates<br>Process of biofertilizer production and use<br>Study of biogas plant setup and process of composting<br>Production of Value-Added Products Through Bioprocessing Technology<br>Isolation, production and purification of industrial enzymes<br>Process of production of organic disinfectants from plants<br>Production of novel fermented food products |

**UNIT-II**

Foundation of Entrepreneurial Product Development and Analysis
Introduction to Entrepreneurship, types and functions
Idea generation, prototype development and idea evaluation
Analysis and shelf-life study
Cost Calculation and project report preparation

From Seed to Success: Startup Incubation Journey
Ideation to business model development
Startups and types, Sources of funding
Building a business model/plan: concept, format and components
Introduction to Incubation: pre-incubation phase, incubation phase and acceleration phase
Go-to Market strategy: Effective ways of marketing and branding for start-up.

11 **Suggested Resources for Reading**


1. Miles, P.G., & Chang, S.-T. (2004). Mushrooms: Cultivation, Nutritional Value, Medicinal Effect, and Environmental Impact (2nd ed.). CRC Press.
2. Desai,V. (2007). Dynamics of Entrepreneurial Development and Management. Himalaya Publishing House.
3. Sen, R., & Roy, S. (2022). Biofuel Production: Biological Technologies and Methodologies (1st ed.). CRC Press.
4. Agarwal S, Kumari S, Khan S. (2021). Bio-entrepreneurship and Transferring Technology into Product Development (pp.1-21) Publisher: IGI GLOBAL
5. Phan PH, Mian SA, Lamine W. (2016). Technology Entrepreneurship and Business Incubation: Theory, Practice, Lessons Learned. World Scientific.


**Note:** Latest edition of text books may be used.

| | COURSE TITLE | BASICS OF FASHION ORNAMENTATION |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 203 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 6 hours/week of practical = 90 hours of practical<br>(1 credit = 30 hours of practical based on 6 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits |
| 5 | **Max. Marks and Time** | 75 Marks (60 PR + 15 IA)<br>i. The exam of this course will be of 3hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. |
| 6 | **Course Objectives** | 1. To develop the skill in various types of embroidery stitches and other methods of ornamentation and use them effectively in designing the garments.<br>2. To beautify the dress by using different material.<br>3. To give an insight about the Indian Heritage-embroideries and textiles of different states. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br>CO1. Can work as Textile designer.<br>CO2. Can work in Fashion Industry.<br>CO3. Can start own entrepreneur.<br>CO4. Can join/run a Fashion house. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into TWO units:**<br>For practical exam, there should be at least two questions for practical performance of 50 marks and one written part of maximum 10 marks. |
| 10 | **Course Content** | |

**UNIT-I**

Study of basic hand embroidery stitches: Flat – Running, Back, Stem. Satin, long and short; Looped – Chain, Feather, Button hole, Lazy daisy; Cross – Simple cross, Herringbone; Knotted – French Knot and Bullion Knot

Basic techniques of Tie and Dye: Understanding of basic sewing techniques of ornamentation – Tucks, Pleats, Gathers, Shirring, Ruffles, Piping

**UNIT-II**

Sketching of different types of motifs.
Samples of Stencil and Block Printing.
Fabric painting – Traced designs and free hand
Introduction to Fashion Maker machines and fancy stitches**.**

11 **Suggested Resources for Reading**


1. Nordan Mary., “Decorative Embroidery” Reader’s Digest Association, New York, 1997.
2. Singer, Decorative Machine Stitching, CyDeCosse Incorporated Publisher, Minnesota 1990.
3. Jane Bolsover, Sewing School Basics, CICO Books, London, 2014.


**Note:** Latest edition of text books may be used.

| | COURSE TITLE | FRONT OFFICE MANAGEMENT |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 204 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 30 hours of lectures for Theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits [2+1] |
| 5 | **Max. Marks and Time** | 75 Marks<br><br>**Ext (Th.)** \| **Int (Th.)** \| **Ext (Pr.)** \| **Int (Pr.)**<br>40 \| 10 \| 20 \| 5<br><br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. |
| 6 | **Course Objectives** | This course is designed to equip students with the necessary skills and knowledge to effectively manage front office operations in various industries. Emphasis will be placed on customer service, communication, and organizational skills required for successful front office management. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br><br>CO1. Handle Front Office equipment and manage the guest during check in and check out<br>CO2. Register and reserve guest at hotel front office<br>CO3. Implement hotel safety and security procedures and maintain quality guest service |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into TWO units**:<br><br>**Note**: The question paper of each subject covering the entire course shall be divided into three sections<br>**Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks.<br>**Section B** (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.<br>**Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. |

10 **Course Content**

**UNIT-I**

Introduction to Front Office Management: Definition and scope of front office management; Importance of the front office in business operations; Role of technology in front office management; Communication skills for front office professionals, Importance of inter-departmental communication; Customer service principles in front office operations, Customer relationship management philosophy, Customer handling and satisfactionmanagement

Front Office Operations: Managing reservations and bookings; Check-in and check-out procedures; Handling customer inquiries and complaints; Coordinating with other departments; Front office security and safetymeasures, Personal care and safety

**UNIT-II**

Personnel Management and Professionalism in Front Office: Recruitment and training of front office staff; Performance evaluation and motivation; Team buildingand conflict resolution; Time management for front office personnel, Importance of personality development in front office roles, Professionalism in front office management, Enhancing personal and professional growth, Dress code and appearance standards in front office roles, Effective body language for positive guest interactions, Customer etiquette, Social etiquette, Corporate etiquette

Technology in Front Office Management: Use of technology in reservation systems; Front office software and applications; Data security andprivacy considerations; Integration with other departments; Emerging trends in front office technology.

11 **Suggested Resources for Reading**


1. James A. Bardi, “Hotel Front Office Management,” Wiley, 2012.
2. Jatashankar R. Tewari, “Front Office Management in Hospitality Lodging Operations,” Delmar Cengage Learning, 2006.
3. Peter Abbott, “Front Office: Procedures, Social Skills, Yield, and Management Control,” Butterworth-Heinemann, 2008.
4. Ahmed Ismail, “Front Office Management,” Lambert Academic Publishing, 2011.
5. Sudhir Andrews, “Front Office Management,” Tata McGraw-Hill Education, 2009**.**


**Note**: Latest edition of text books may be used.

<table>
<tr><th></th><th>COURSE TITLE</th><th>INTRODUCTION TO ECONOMETRICS</th></tr>
<tr><td>1</td><td>Course Code</td><td>B.Voc. SEC 205</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Skill Enhancement Courses (SEC)</td></tr>
<tr><td>3</td><td>Course Duration/ LP (Lectures, Practical)</td><td>30 hours of lectures for Theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>3 credits [2+1]</td></tr>
<tr><td>5</td><td>Max. Marks and Time</td><td>75 Marks<br><table><tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr><tr><td>40</td><td>10</td><td>20</td><td>5</td></tr></table><br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.<br>iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>6</td><td>Course Objectives</td><td>To familiarize the students with the economic application related to econometrics.</td></tr>
<tr><td>7</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to:<br>CO1. To train the students in applying the methodology of various concepts in software<br>CO2. To impart practical knowledge about the skill of analysing and interpreting the data</td></tr>
<tr><td>8</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>9</td><td>Instructions for paper setters and candidates</td><td>The syllabus has been divided into TWO units:<br>Note: The question paper of each subject covering the entire course shall be divided into three sections<br>Section A (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks.<br>Section B (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.<br>Section C (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.</td></tr>
</table>

10 **Course Content**

**UNIT-I**

Introduction: Meaning, definition, nature, goals, scope, and limitation of econometrics, Methodology of Economic Research.

Simple Regression Models: Assumptions, Estimation of two variable models by method of Ordinary Least squares, Properties of two variable models (numerical and statistical properties), Goodness of fit-R Square & Adjusted R Square.

*Practical: Estimation of two variables Regression Model and Analysis of R square and adjusted R square

**UNIT-II**

Testing of hypothesis: Basic Concepts, types of hypotheses, Type I, Type II errors, Procedure of hypothesis testing, t test and f test.

Multiple Regression models: Assumptions, Estimation of model by method of Ordinary Least Squares, Properties of the Least Squares estimators, the Gauss-Markov theorem.

*Practical: Estimation and Analysis of Multiple Regression Model using SPSS.

Technology In Front Office Management: Use of technology in reservation systems; Front office software and applications; Data security andprivacy considerations; Integration with other departments; Emerging trends in front office technology.

11 **Suggested Resources for Reading**


1. D. N. Gujarati and D.C. Porter (2009). Essentials of Econometrics (4thed.) McGraw Hill, International Edition.
2. Christopher Dougherty (2007). Introduction to Econometrics (3rded.) Oxford University Press, Indian edition.
3. Jan Kmenta, (2008). Elements of Econometrics. (2nded.). Indian Reprint, Khosla Publishing House
4. Richard J. Larsen and Morris L. Marx (2011). An Introduction to Mathematical Statistics and its Applications. Prentice Hall.
5. Ohnson, J. (1977): Econometrics Methods, 2ndEdition, McGraw Hill, New
6. Intrilligator, M. D. (1978): Econometric Models and Applications, Prentice Hall
7. Kendall, M. G. &A. Sturat (eds.): Advanced Theory of Statistics, Vols. I & II, Griffin and Co., London.
8. Maddala, G. S. (1977): Econometrics, New Delhi, McGraw Hill.
9. Wallis, K. F. (1973): Topics in Applied Econometrics, London, Bray Hills


**Note**: Latest edition of text books may be used.

| | COURSE TITLE | GST RETURN FILING |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 206 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 30 hours of lectures for Theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits |
| 5 | **Max. Marks and Time** | 75 Marks (60 PR+ 15 IA)<br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.<br>iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees |
| 6 | **Course Objectives** | To acquaint the students with basics of GST and also the process of creating GST account and filing GST return as per the law and rules. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br>CO1. Develop a detailed understanding of documents, records, books of accounts and tax returns, and the requirements for the payment of Goods and Services Tax.<br>CO2. Carry out registration of companies under GST.<br>CO3. Compute tax liability under GST.<br>CO4. File returns and maintain records of the same for audit purpose.<br>CO5. Provide expert advice to assesses under GST. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into two units:**<br>**Note:** The question paper of each subject covering the entire course shall be divided into three sections<br>**Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks.<br>**Section B** (15 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.<br>**Section C** (15 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. |

10 **Course Content**

**UNIT-I**

Goods & Services with their cross linkages, Fundamental Concepts of GST, Levy of IGST, identify whether a transaction is taxable under CGST, IGST or SGST, Incidence of Taxation, Time of Supply of Goods, Place of supply, Location of supplier of goods, Registration under GST

**UNIT-II**

GST slabs and items under them, Define Input Credit, Identifying Input Tax Credit eligibility, maintaining different types of ledgers, preparing documents such as Invoice, Credit Note and Debit Note, identifying types of Payment, Modes of Payment, Distinguish between TDS and TCS, marketing oneself as GST expert.

11 **Suggested Resources for Reading**


1. Ahuja, Girish and Gupta, Ravi. Systematic Approach to GST. Wolters Kluwer
2. Sekhon, Shailinder., GST. Unlocking the complexities of Indirect taxes. Published by Sumirat Publication and Bookman, New Delhi
3. Bansal, K. M., GST & Customs Law, Taxman Publication
4. Datey, V. S. All about GST, A Complete Guide to Model GST Law. Taxman Publication
5. Batra, A., Batra, A., and Jaggi, R. Bharat's GST Ready Reckoner. Aggarwal Law House.
6. Taxmann's GST Manual with GST Law Guide and Digest of Landmark Rulings. Taxmann Publications.


**Note:** Latest edition of text books may be used.

# SEMESTER III

| S. No. | Course Code | Course Title |
|---|---|---|
| 1. | B.Voc. SEC 301 | Analytical Skills on Benchtop Spectroscopic Techniques |
| 2. | B.Voc. SEC 302 | MATLAB in Numerical Analysis |
| 3. | B.Voc. SEC 303 | AutoCAD for Designing |
| 4. | B.Voc. SEC 304 | Mathematical Modelling in Biology |
| 5. | B.Voc. SEC 305 | Entrepreneurship in Millet Processing |
| 6. | B.Voc. SEC 306 | Computational Physics |
| 7. | B.Voc. SEC 307 | Digital Art Using Corel Draw |
| 8. | B.Voc. SEC 308 | Negotiation and Conflict Management |
| 9. | B.Voc. SEC 309 | Project Management |

| | COURSE TITLE | ANALYTICAL SKILLS ON BENCHTOP SPECTROSCOPIC TECHNIQUES |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 301 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 30 hours of lectures for Theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits |
| 5 | **Max. Marks and Time** | 75 Marks<br><br>**Ext (Th.)** \| **Int (Th.)** \| **Ext (Pr.)** \| **Int (Pr.)**<br>40 \| 10 \| 20 \| 5<br><br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.<br>iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/ College to the examinees. |
| 6 | **Course Objectives** | i. To introduce students to the basic principles of spectroscopy.<br>ii. To familiarize students with the theoretical foundations and practical applications of NMR, IR, UV-Vis, and Fluorescence spectroscopy.<br>iii. To provide students with hands-on experience in operating spectroscopic instruments and interpreting spectroscopic data.<br>iv. To enable students to apply spectroscopic techniques in chemical analysis and research. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br><br>CO1. Handle and operate basic spectroscopic instruments and software.<br>CO2. Apply the spectroscopic skills for the structure elucidation of unknown simple organic and inorganic compounds.<br>CO3. Detect the concentration of various ions in aqueous and non-aqueous solutions.<br>CO4. Collect, process and interpret the data from spectroscopic instruments.<br>CO5. Apply the spectroscopic skills in quality analysis and quality control sections in chemical industries. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into TWO units.**<br><br>**Note:** The question paper of each subject covering the entire course shall be divided into three sections<br>**Section A** (8 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2 marks; the total weightage being 8 marks.<br>**Section B** (16 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two |

questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks.

**Section C (**16 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks. Will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks.

10 **Course Content**

**UNIT-I**

Introduction to Spectroscopy: Definition and significance of spectroscopy, Historical developments, Electromagnetic Radiation and Molecular Interactions, Nature of electromagnetic radiation, Interaction of radiation with matter

Ultraviolet-Visible (UV-Vis) Spectroscopy: Principles of UV-Vis spectroscopy, Working of UV-Visible Spectrophotometer Beer-Lambert Law and its applications, UV-Vis spectra interpretation

Fluorescence Spectroscopy: Principles of fluorescence spectroscopy, Working of Fluorescence Spectrophotometer, Fluorescence, Phosphorescence, Jablonski diagram, Fluorescence spectra interpretation and applications

**UNIT-II**

Infrared (IR) Spectroscopy: Principles of IR spectroscopy, Working of IR Spectrotometer, Vibrational modes in molecules, IR spectra interpretation and applications

Nuclear Magnetic Resonance (NMR) Spectroscopy: Principles of NMR spectroscopy, Working of NMR Spectrotometer Nuclear spin and resonance, Chemical shift, Shielding, deshielding and coupling constants, NMR spectra interpretation and applications

**Practical Work:**

Introduction to Laboratory Safety and Instrumentation

(i) Safety guidelines in the laboratory
(ii) Introduction to spectroscopic instruments

UV-Visible (UV-Vis) Spectroscopy Experiment

(i) Stock and standard solution preparation techniques
(ii) Operation of UV-Vis spectrophotometer
(iii) Recording and Interpretation of UV-Visible spectra
(iv) Analysis of results

Experiment 1. Spectrophotometric determination of concentration of Fe (III) ions in given Ferric Chloride solution using thiocyanate method.
Experiment 2. Spectrophotometric determination of concentration of Cu (II) ions in given Copper Sulphate solution using Ammonia solution.

Fluorescence Spectroscopy Experiment

(i) Sample and buffer preparation techniques
(ii) Calibration of Fluorescent Probes(dyes)
(iii) Operation of fluorescence spectrophotometer
(iv) Measurement and interpretation of fluorescence spectra
(v) Analysis of results

Experiment 7. Investigating the effect of pH on Fluorescence Intensity of Fluorescein Solution.
Experiment 8. To assess contaminants levels in industrial water sample using Fluorescence Spectroscopy.

Infrared (IR) Spectroscopy Experiment

(i) Sample handling and preparation
(ii) Operation of Bench top IR spectrometer
(iii) Recording and interpretation of IR spectra
(iv) Analysis of results

Experiment 3. Interpretation of the functional groups in the given organic compounds using IR spectrometer – Acetanilide, Benzaldehyde, Naphthylamine, and Iso-propanol

Experiment 4. Interpretation of the functional groups in the given organic compounds using IR

11 **Suggested Resources for Reading**


1. Pass, G.; Sutcliffe Practical Inorganic Chemistry, 1st edition, Chapmann and Hall Ltd., 1968.
2. Vogel, A.I. Text Book of Practical Organic Chemistry, ELBS, 5th edition, Longman
3. Jeffery G H; Bassett J; Mendham J.; Denney R C Vogel's Textbook of Quantitative Chemical Analysis, 5th Edition, Publisher: Longman Scientific Technical, Co-published in the United States with John Wiley Sons, Inc., New York


**Note**: Latest edition of text books may be used.

| | COURSE TITLE | MATLAB IN NUMERICAL ANALYSIS |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 302 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 90 hours of practicals<br>(1 credit = 30 hours of practical based on 6 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits |
| 5 | **Max. Marks and Time** | 75 Marks (60 PR + 15 IA)<br>i. The exam of this course will be of 3hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. |
| 6 | **Course Objectives** | The objective of this course is to provide exposure to the students regarding the application of MATLAB software in numerical methods. The course will cover the basics of MATLAB and introduce its usage in solving the numerical problems which include root finding of an equation, errors, approximations, plotting, curve fittings, solving the problems of differentiation, integration and ODE'. |
| 7 | **Course Outcomes** | After completion of this course, the student should be able to understand the most significant concepts and commands in MATLAB which are used for solving mathematical problems of approximations occurring in modern engineering and science. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | For practical exam, there should be at least two questions for practical performance of 50 marks and one written part of maximum 10 marks. |
| 10 | **Course Content** | |

**UNIT-I**

Basics of MATLAB programming: Array operations in MATLAB, Loops and execution control, working with scripts and functions, Plotting and program output

Polynomials: Finding Roots of a polynomial, Operations of polynomials: Addition, Multiplication, Division, Derivatives, Curve fitting with polynomials

Errors and approximations: Using Taylor's/ Maclaurin series: Truncation and round-off errors, error propagation, global and truncation errors

Root finding of nonlinear equations: Bisection method, Fixed point-iteration method, Regula-Falsi, Secant method, Newton Raphson method

**UNIT-II**

Solving a system of linear equations: Gauss elimination method, Gauss Jordan method, LU Decomposition method, inverse of a matrix, Iterative methods: Gauss – Jacobi, Gauss-Seidal methods, Matrix eigenvalues and eigenvectors; Power method

Curve Fitting and interpolation techniques: Interpolation using a single polynomial, Lagrange and Newton's polynomials, Piecewise interpolation, linear, quadratic, and cubic splines, use of MATLAB built-in functions for curve fitting and interpolation

Numerical differentiation and integration: Differentiation using Lagrange polynomials, use of MATLAB built-in functions for numerical differentiation, Integration using Rectangle and midpoint methods, trapezoidal method, Simpson's methods, use of MATLAB built-in functions for integration

Solving ODE initial value problems: Runge-Kutta methods, multistep methods, predictor-corrector methods, system of first-order ODEs, higher-order IVP; local truncation error in 2nd-order Runge-Kutta method, step size for desired accuracy.

11 **Suggested Resources for Reading**


1. Chapra, Steven C. (2018). Applied Numerical Methods with MATLAB for Engineers and Scientists (4th ed.). McGraw-Hill Education.
2. Fausett, Laurene V. (2009). Applied Numerical Analysis Using MATLAB. Pearson. India.
3. Jain, M. K., Iyengar, S. R. K., & Jain R. K. (2012). Numerical Methods for Scientific and Engineering Computation (6th ed.). New Age International Publishers. Delhi


**Note**: Latest edition of text books may be used.

| | COURSE TITLE | AUTOCAD FOR DESIGNING |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 303 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 30 hours of lectures for Theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits |
| 5 | **Max. Marks and Time** | 75 Marks<br>**Ext (Th.)** \| **Int (Th.)** \| **Ext (Pr.)** \| **Int (Pr.)**<br>40 \| 10 \| 20 \| 5<br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. |
| 6 | **Course Objectives** | The objective of this course is to familiarize students with the fundamentals of AutoCAD for designing tools, objects and layouts. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br>CO1. Using AutoCAD as a drafting and design tool for designing common products.<br>CO2. Apply basic CAD concepts to develop and construct accurate 2D geometrical objects.<br>CO3. Draft, manipulate and modify 2D drawings and figures.<br>CO4. Apply elements of drafting such as layers, dimensions, drawing formats, and 2D figures in projects. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into two units:**<br>**Note:** The question paper of each subject covering the entire course shall be divided into three sections<br>**Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks.<br>**Section B** (15 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.<br>**Section C** (15 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. |

10 **Course Content**

**UNIT-I**

Creating, Opening & Saving Drawings, Exploring the AutoCAD interface, Zooming and Panning, Polar and Ortho Tracking, Entering Coordinates and Angles, Object Snaps and Tracking, Using the Mouse, Keyboard, and Enter Key to run AutoCAD commands.

Drawing- Lines, Circles, Rectangles, Polylines, Arcs, Polygons, Ellipses, Layer States, Properties by Layer, Layer Tools. Properties Palette, Quick Select, Select Similar, Measure Geometry Tools. Erasing parts or complete drawing.

**UNIT-II**

Basic Editing Commands- Move, Copy, Rotate, Mirror, Scale, Using the reference option with the Scale Tool, Advanced Editing Commands Trim, Extend, Fillet and Chamfer, Polyline, Edit and Spline Offset, Explode and Join.

Insert Block Command- Inserting Blocks with Tool Palette, Dynamic Blocks. Adding Annotations with single line and Multiline text tool, Dimensioning Tool, Dimension Styles, Editing Dimensions, Hatching - Hatch Command, Hatch Editor Ribbon, Tab Saving and Applying Hatches with Tool Palettes.

11 **Suggested Resources for Reading**

1. Autodesk AutoCAD 2021, Learn CAD With Ease (For Beginners)
2. AutoCAD Reference Guide Book Paperback – 1 January 2020 by CAD Desk

**Note:** Latest edition of text books may be used.

<table>
<tr><th></th><th>COURSE TITLE</th><th>MATHEMATICAL MODELLING IN BIOLOGY</th></tr>
<tr><td>1</td><td>Course Code</td><td>B.Voc. SEC 304</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Skill Enhancement Courses (SEC)</td></tr>
<tr><td>3</td><td>Course Duration/ LP (Lectures, Practical)</td><td>30 hours of lectures for theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>3 credits</td></tr>
<tr><td>5</td><td>Max. Marks and Time</td><td>75 Marks
<table>
<tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr>
<tr><td>40</td><td>10</td><td>20</td><td>5</td></tr>
</table>
i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.<br>iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>6</td><td>Course Objectives</td><td>The ability to build and evaluate models and/or interpret model outputs is a key requirement for future scientists, biologists, industry leaders, resource managers, and policy makers. Keeping in mind the aforementioned point the objective of this course is to train students in techniques of model development, their use and assessment. Adding to this, also to teach students to apply physical sciences perspectives to biological research.</td></tr>
<tr><td>7</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to:<br>CO1. Represent real-world systems from science and technology in a mathematical framework.<br>CO2. Construct mathematical models for biological systems.<br>CO3. Draw inferences from models using quantitative reasoning, and exploration using multiple representations such as equations, plots, tables and graphs.<br>CO4. Apply critical thinking to address problems in an interdisciplinary group setting.</td></tr>
<tr><td>8</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>9</td><td>Instructions for paper setters and candidates</td><td>The syllabus has been divided into two units:<br>Note: The question paper of each subject covering the entire course shall be divided into three sections<br>Section A (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks.<br>Section B (15 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.</td></tr>
</table>

**Section C** (15 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.

10 **Course Content**

**UNIT-I**

Mathematical models: Introduction, techniques and classifications. Mathematical modelling through ODE of first order. Application of first order to acceleration-velocity model, growth and decay model. Population dynamics: discrete and continuous models, aging structure, interacting species (competition, predation, parasitism).

Introduction to compartmental models, Infectious disease modeling: SIR/SEIR (persistence, eradication, control), metapopulation models; Vector-borne diseases; Macro-parasite diseases; Individual- based models population growth models, prey-predator model, drug assimilation into blood. Exponential growth of population, limited growth of population, limited growth with harvesting.

*Practical: Test your knowledge: basic definitions, modeling scales, classification.

**UNIT-II**

Ordinary differential equations and systems: Linear systems (eigenvalue method). Equilibria, stability, Bifurcation (Routh-Hurwitz; periodic orbits, stable/ unstable orbits). Poincare-Bendixon theorem.*

Discrete dynamical systems, Periodic points: stability, hyperbolicity, linearization, Bifurcations. Mathematical modelling of biological phenomenon through difference equations: theory and solutions. Application and analysis of difference equations to population dynamics.*

*Practical: Formulation/writing of the model/code using MATLAB software. Plotting the system using MATLAB/Gnuplot. Simulation of various models to be done on MATLAB/R and XPPAUT.

11 **Suggested Resources for Reading**


1. S. L. Ross, Differential Equations, 3rd Edition, John Wiley and Sons, India.
2. J.N. Kapur: Mathematical modeling, Wiley Eastern limited, 1990.
2. J.N. Kapur. Mathematical models in biology and medicine. Affiliated East-West Press, 1985.
3. N.H. Ibragimov. Practical Course in Differential Equations and Mathematical Modelling, A: Classical and New Methods. Nonlinear Mathematical Models. Symmetry and Invariance Principles. World Scientific Publishing Company, 2009.
4. B. Barnes and G. R. Fulford. Mathematical modelling with case studies: using maple and MATLAB. Vol. 25. CRC Press, 2014.


**Note:** Latest edition of text books may be used.

| | COURSE TITLE | ENTREPRENEURSHIP IN MILLET PROCESSING |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 305 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 90 hours of practicals<br>(1 credit = 15 hours of theory learning based on 3 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits |
| 5 | **Max. Marks and Time** | 75 Marks (60 PR + 15 IA)<br>i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. |
| 6 | **Course Objectives** | The purpose of this course is to provide comprehensive information about millets, which are the ancient and forgotten golden grains, and to learn the commercial potential associated with this emerging topic. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to<br>CO1. Learn about millets as functional foods and their potential benefits.<br>CO2. Gain knowledge about nutritious and delicious dishes of millet in detail to incorporate into our regular diet and about the environmental benefits of millets.<br>CO3. Learn commercial aspects and requirements of starting millet-based business or start-up. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into TWO units:**<br>**Note:** The question paper of each subject covering the entire course shall be divided into three sections<br>For practical exam, there should be at least two questions for practical performance of 50 marks and one written part of maximum 10 marks. |
| 10 | **Course Content** | **UNIT-I**<br>Introduction to millets: types & Nutritional Benefits<br>Definition of millets.<br>Types & Varieties<br>Positive & Neutral Millets<br>Why include millet in our diet?<br>Identifying polished & unpolished millets?<br>How to store millets?<br>Health benefits of Indian millets<br>Environmental benefits of millets |

Introduction to Entrepreneurship:
Ways of sensing opportunities.
Idea generation.
SWOT analysis.
Steps of Product Development
Analysis and shelf-life study
Cost Calculation and project report preparation

**UNIT-II**

Traditional Foods Based on Major Millets:
Sorghum
Pearl Millet
Value addition from by-products:
Sorghum bran peda (sweet)
Sorghum based energy bars
Sorghum bran fryums

Traditional Foods Based on Minor Millets:
Finger Millet
Proso Millet
Kodo Millet
Barnyard Millet
Foxtail Millet

Business opportunities in various segments:
Market analysis, demand for millet-based products
Millets in food industries, farmer producer company
Millet Primary Processing Unit
Millets Value Addition
Packaging and Labelling requirements for millets-based products

11 **Suggested Resources for Reading**


1. Yadav, Latika & Upasana, Upasana. (2024). Millets - Rediscover Ancient Grains.
2. Kumar Siroha, A., Punia, S., Singh Purewal, S., & Singh Sandhu, K. (2021). Millets: Properties, Processing, and Health Benefits (1st ed.). CRC Press.
3. Kumar, A., Tomer, V., Kumar, M., & Chawla, P. (2024). Millets: Cultivation, Processing, and Utilization (1st ed.). CRC Press.
4. Bairwa, Mahesh & Kumari, Rajni & Semwal, Rajeev. (2023). THE MIRACLES OF MILLETS (A Tribute to International Year of Millets 2023).


**Note**: Latest edition of text books may be used.

| | COURSE TITLE | COMPUTATIONAL PHYSICS |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 306 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 15 hours of lectures for Theory + 60 hours of practicals<br>(1 credit = 15 hours of theory learning based on 1 hrs./week)<br>(1 credit = 30 hours of practical based on 4 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits [1+2] |
| 5 | **Max. Marks and Time** | 75 Marks<br>**Ext (Th.)** 20 \| **Int (Th.)** 5 \| **Ext (Pr.)** 40 \| **Int (Pr.)** 10<br>i. The exam of this course will be of 3hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.<br>iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees. |
| 6 | **Course Objectives** | 1. Understanding Basic concepts of C and C++ Programming language, solving and analyzing Physics concepts<br>2. Training of Arduino programming.<br>3. Training about preparing working models using physics concepts and Arduino programming. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br>CO1. Design working models based on physics concepts and programming to make them skill oriented with practical amenabilities.<br>CO2. Making scientific skills innovative and interesting to learn and understand.<br>CO3. Understanding the significance of Artificial Intelligence (AI) through scientific working models and Arduino Programming.<br>CO4. Utilize Arduino Programming for making automatic working models from manual research projects.<br>CO5. Will develop critical thinking, design thinking and innovative research paradigm approach for science. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into TWO units**.<br>**Note:** The question paper of each subject covering the entire course shall be divided into three sections<br>**Section A** (4 marks) This section will have 4 short answer questions from the entire syllabus. Students are required to attempt 2 questions from this section. Each question will carry 2 marks; the total weightage being 4 marks.<br>**Section B** (8 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks.<br>**Section C** (8 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt |

two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks.

10 **Course Content**

**UNIT-I**

Basics and concepts of C and C++ Programming language, solving and analyzing physics concepts: Variable and data types in C++, Expressions, Classes, Objects, Operators, Control statements, Functions, Types of Functions, Function arguments, Function overloading, Pointers and Arrays, Strings, Structures, Constructors and Destructors, Inheritance, Streams and File Handling, Numerical techniques using Runga Kutta Method, Simpson rule and other numerical analytical techniques. Understanding and analyzing Radioactive disintegrations, random numbers generations, Fourier transforms, Rise and decay, atomic nuclei and other physics related programs using C++ programming. Training students to prepare algorithms and compiling Computational Physics programs.

**UNIT-II**

Arduino programming: Basics of electronics, programming and coding. Programming codes for Integrated Circuit (IC) boards and utilizing various sensors like infrared sensors, ultrasonic sensors, UV sensor and many more to apply physics experimentally.

Training about preparing working model using physics concepts and Arduino programming: Training students to design working models based on physics concepts and programming to make them skill oriented with practical amenabilities.

11 **Suggested Resources for Reading**


1. The C++ Programming Language (4th Edition) By Bjarne Stroustrup
2. Introduction to Object Oriented Programming, By Balaguru Swamy
3. Arduino Workshop: A Hands-On Introduction with 65 Projects 1st Edition.
4. Arduino: Advanced Methods and Strategies of Using Arduino.
5. Practical C++ Programming (2nd Edition) By Steve Oualline
6. Effective Modern C++ (2014) By Scott Meyers
7. Arduino Robotics (Technology in Action).


**Note**: Latest edition of text books may be used.

| | COURSE TITLE | DIGITAL ART USING CORELDRAW |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 307 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 15 hours of lectures for Theory + 60 hours of practicals<br>(1 credit = 15 hours of theory learning based on 1 hrs./week)<br>(1 credit = 30 hours of practical based on 4 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit |
| 4 | **Credits** | 3 credits [1+2] |
| 5 | **Max. Marks and Time** | 75 Marks<br><br>**Ext (Th.)** \| **Int (Th.)** \| **Ext (Pr.)** \| **Int (Pr.)**<br>20 \| 5 \| 40 \| 10<br><br>i. The exam of this course will be of 3hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. |
| 6 | **Course Objectives** | To familiarize the students with working of CorelDRAW, as an important graphics editor. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to:<br><br>CO1. To help them use various tools of CorelDRAW.<br>CO2. To inculcate skills for creating brochures, newsletters, business card etc.<br>CO3. Skilled for primarily for marketing and advertising businesses, particularly those that specialize in print advertising development. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into TWO units**.<br><br>**Note:** The question paper of each subject covering the entire course shall be divided into three sections<br>**Section A** (4 marks) This section will have 4 short answer questions from the entire syllabus. Students are required to attempt 2 questions from this section. Each question will carry 2 marks; the total weightage being 4 marks.<br>**Section B** (8 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks.<br>**Section C** (8 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks. |

10 **Course Content**

**UNIT-I**

Introduction: Introduction to CorelDRAW, document info, working with toolbars, working with docker windows, using views, zoom, workspaces. Customizing Options: Using Multiple workspaces, Customizing the Toolbars, Using Shortcuts, Saving Defaults, Setting File Backups.

Basic Drawing: Selection techniques, objects locking, digger selection, working with: lines, freehand curves, ellipses, rectangles, grids, rulers, guidelines, and snapping to objects.

**UNIT-II**

Advanced Drawing: Working with curves, polygons, manipulating objects, working with nodes: adding, deleting, rotating and skewing.

Styles and Templates: Introduction to styles and templates, creating a style, applying a style, copying properties, use of templates.

11 **Suggested Resources for Reading**


1. CorelDRAW in easy steps by Stepman 2009
2. Teach Yourself CorelDRAW by Karlins David 2009
3. CorelDRAW 5, 9, 12 for beginners by David Gray
4. Computer Graphics by Donald Hearn and M. Pauline Baker, PHI, 2009.
5. Mastering CorelDRAW by Altman, BPB. 2009


**Note**: Latest edition of text books may be used.

<table>
<tr><th></th><th>COURSE TITLE</th><th>NEGOTIATION AND CONFLICT MANAGEMENT</th></tr>
<tr><td>1</td><td>Course Code</td><td>B.Voc. SEC 308</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Skill Enhancement Courses (SEC)</td></tr>
<tr><td>3</td><td>Course Duration/ LP (Lectures, Practical)</td><td>30 hours of lectures for Theory + 30 hours of practicals<br>(1 credit = 15 hours of theory learning based on 2 hrs./week)<br>(1 credit = 30 hours of practical based on 2 hrs./week)<br>(1 Semester = 15 weeks)<br>3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>3 credits</td></tr>
<tr><td>5</td><td>Max. Marks and Time</td><td>75 Marks
<table>
<tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr>
<tr><td>40</td><td>10</td><td>20</td><td>5</td></tr>
</table>
i. The exam of this course will be of 3 hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.</td></tr>
<tr><td>6</td><td>Course Objectives</td><td>To make the students understand dynamics of conflict in organizations and to build their capacity to prevent and resolve conflicts in an amicable manner.</td></tr>
<tr><td>7</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to:<br><br>CO1. Understand the nature of conflict, including its causes, manifestations, and consequences.<br>CO2. Analyze different conflict resolution approaches and their applicability in various contexts, such as interpersonal, organizational, and societal levels.<br>CO3. Evaluate the role of personal and cultural biases, power dynamics, interests, and values in conflict situations and their impact on resolution processes.<br>CO4. Demonstrate proficiency in conflict assessment and analysis tools to diagnose conflict issues and develop intervention strategies.<br>CO5. Explore the ethical dimensions of conflict management, including principles of fairness, justice, and respect for human rights.<br>CO6. Develop effective communication skills for managing and de-escalating conflicts, including active listening, empathy, and assertiveness.</td></tr>
<tr><td>8</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>9</td><td>Instructions for paper setters and candidates</td><td>The syllabus has been divided into TWO units:<br><br>Note: The question paper of each subject covering the entire course shall be divided into three sections<br>Section A (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks.<br>Section B (15 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.<br>Section C (15 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt</td></tr>
</table>

two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.

10 **Course Content**

**UNIT-I**

Conflict: Meaning in context of Human Nature, Causes (Personal, Departmental, Systemic), Types (Functional/Dysfunctional, Short term/Long term, Direct/Indirect)

Conflict Management: Need to Prevent and Resolve Conflicts (Costs and Benefits); Factors and Processes that affect Conflict Management in an organization/situation (Nature, Structure of Authority and Responsibility, Mechanisms for Prevention and Resolution and their use, Leadership, Prominent Values, Communication Network, Stakeholders).

**UNIT-II**

Stages of Conflict Management: Diagnosis the Conflict and its Causes, Understanding take of Parties Involved, Assessment of Potential Costs and Benefits of unresolved Conflict, Identify Strategy of Resolution, Planning implementation of Solution and Follow up.

Strategies/ Approaches to Conflict Management: Avoiding/Prevention, Competition, Accommodation, Compromise, Collaboration, Negotiation, Arbitration, Alternative Dispute Resolution.

11 **Suggested Resources for Reading**


1. DeChurch, Leslie A., and Michelle A. Marks. "Maximizing the benefits of task conflict: The role of conflict management." International Journal of Conflict Management 12.1 (2001): 4-22.
2. Elgoibar, Patricia, Martin Euwema, and Lourdes Munduate. "Conflict Management." Oxford Research Encyclopedia of Psychology. 2017.
2. Himes, Joseph S. Conflict and conflict management. University of Georgia Press, 2008.
3. Thakore, Digvijaysinh. "Conflict and conflict management." IOSR Journal of Business and Management 8.6 (2013): 07-16.
4. Verma, Vijay K. "Conflict management." The Project Management Institute: Project Management Handbook (1998): 353-364.


**Note:** Latest edition of text books may be used.

| | COURSE TITLE | PROJECT MANAGEMENT |
|---|---|---|
| 1 | **Course Code** | B.Voc. SEC 309 |
| 2 | **Typology of Course** | Skill Enhancement Courses (SEC) |
| 3 | **Course Duration/ LP (Lectures, Practical)** | 45 hours of lectures for Theory<br>(1 credit = 15 hours of theory learning based on 3 hrs./week)<br>(1 Semester = 15 weeks) |
| 4 | **Credits** | 3 credits |
| 5 | **Max. Marks and Time** | 75 Marks (60 TH + 15 IA)<br>i. The exam of this course will be of 3hours duration.<br>ii. There will be no objective type questions.<br>iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.<br>iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees |
| 6 | **Course Objectives** | The objective of the course is to help the student understand the basics of efficiently managing projects. |
| 7 | **Course Outcomes** | On successful completion of the course, the students will be able to<br>CO1. Understand the fundamentals of project management.<br>CO2. Study the essential components of a feasibility study.<br>CO3. Understand the principles and techniques of project cost estimation, budgeting and control.<br>CO4. Study project performance monitoring and control. |
| 8 | **Pedagogy of the Course Work** | Class Room Lecture, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. |
| 9 | **Instructions for paper setters and candidates** | **The syllabus has been divided into TWO units:**<br>**Note**: The question paper of each subject covering the entire course shall be divided into three sections.<br>**Section A** (20 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 5 marks; the total weightage being 20 marks.<br>**Section B** (20 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 10 marks; the total weightage being 20 marks.<br>**Section C** (20 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 10 marks; the total weightage being 20 marks. |

10 **Course Content**

**UNIT-I**

Concept of Project: Meaning, Characteristics, Classification of Projects, Project Life Cycle and its Phases. Project Manager: Roles and Responsibilities, Project Management as a Profession.

Generating and Screening Ideas – Steps, Monitoring the Environment, Scouting for Project Ideas, Preliminarily Screening, Project Rating Index. Feasibility Studies – Technical, Financial, Economic, Social, Legal and Managerial.

**UNIT-II**

Project Appraisal Techniques: Objectives, Types and Methods. Project Risks: Meaning, Types, Measurement of Risk, Decision Tree Analysis (Basic Concepts only).

Project Evaluation – Meaning, Evaluation v/s Appraisal, Objectives, Types of Evaluation, Techniques.

Project Organization and Control – Project Network Analysis (Basic concepts of PERT, CPM, Cost and Time Over Run). Project Reporting: Meaning, Purpose, Process, Requirements of a Good Report, Methods, Principles of Good Reporting System.

11 **Suggested Resources for Reading**


1. Chandra, P., Projects: Planning, Analysis, Selection, Financing, Implementation and Review,
2. McGraw Hill Education.
3. Kerzner, H., Project Management: A Systems Approach to Planning, Scheduling, and Controlling. Wiley.
4. Larson, E. W., & Gray, C. F., Project Management: The Managerial Process. McGraw-Hill Education.
5. Schwalbe, K., Information Technology Project Management, Cengage Learning.
6. Meredith, J. R., & Mantel, S. J., Project Management: A Managerial Approach, Wiley.


**Note**: Latest edition of text books may be used.